



Welcome to A Catholic Introduction to Artificial Intelligence Class 3! AI in the 20th Century

"The question is not whether machines can think, but whether men do." — B. F. Skinner

Learning Objectives

By the end of this class, students will be able to:

1. Describe the evolution of AI from the 1950s through the 1990s.
2. Identify key developments such as ELIZA, expert systems, and robotics.
3. Explain why early AI efforts both excited and disappointed researchers.
4. Recognize the influence of AI on 20th-century science and culture.

1. Class 2 Review

Artificial intelligence has gone through several major shifts since its earliest days, and understanding these changes helps explain both past challenges and today's progress. Early AI researchers in the 1950s and 1960s believed that computers would soon match human reasoning. Key figures such as Alan Turing, John McCarthy, Marvin Minsky, and Herbert Simon laid the foundations for the field, but their predictions often exceeded what technology could realistically achieve at the time.

These overly optimistic expectations led to periods known as *AI Winters*. An AI Winter occurs when excitement and funding collapse because AI systems fail to deliver on the promises made by researchers or the media. This happened more than once: early language-processing systems struggled with real-world complexity, and later expert systems in the 1980s proved too brittle and expensive to maintain. Each time, the gap between hype and actual performance caused disappointment and slowed progress.

A major theme across these winters was the cycle of *overpromising and underdelivering*. Researchers showcased impressive demos in narrow settings, but those systems could not generalize to broader tasks. As limitations became clear, governments and companies withdrew support. This pattern created a repeating rise-and-fall curve in AI enthusiasm, often described as the AI hype cycle.

Today's AI boom is different because modern systems are powered by enormous datasets, far more powerful hardware, and new learning techniques such as deep learning and transformers. Instead of relying on hand-written rules, today's models learn patterns from billions of examples. This combination of data, computing power, and new algorithms allows AI to deliver real, measurable results across language, vision, science, and industry—making another AI Winter less likely, even as researchers remain cautious about repeating past mistakes

2. Key Early AI Programs (1950s - 1960s)

- SNARC (1950)

SNARC (the **Stochastic Neural Analog Reinforcement Calculator**) was an early artificial intelligence machine built around 1950 by Marvin Minsky and Dean Edmonds. It is often considered one of the first attempts to create a system that mimics how the brain learns. Rather than using modern digital software, SNARC was constructed from about 3,000 vacuum tubes and analog components, forming a physical network of roughly 40 artificial "neurons." These neurons were connected in a way that allowed the system to simulate a rat navigating a maze, where successful paths (finding "food") would strengthen the connections that led to that success.

What makes SNARC especially important is that it introduced the core idea of **reinforcement learning** decades before it became a major field in AI. The system improved its behavior through trial and error, reinforcing successful actions—much like how animals learn. Although it was small and impractical by today's standards, SNARC showed that intelligent behavior could emerge from networks of simple units adjusting their connections over time. It represents an early milestone in AI history and highlights that many modern techniques, such as neural networks and learning from rewards, have roots that go back to the very beginning of the field.

- First AI Conference(1956)

John McCarthy of Dartmouth College in NH convince Marvin Minsky and others to attend a 2-month long workshop during the summer of 1956. The proposal stated:

"We propose that a 2 month, 10 man study of artificial intelligence be carried out during the summer of 1956 at Dartmouth College in Hanover, New Hampshire. The study is to proceed on the basis of the conjecture that every aspect of learning or any other feature of intelligence can in principle be so precisely described that a machine can be made to simulate it. An attempt will be made to find how to make machines use language, form abstractions and concepts, solve kinds of problems now reserved for humans, and improve themselves."

No significant breakthroughs occurred during this study.

- General Problem Solver (late 1950s)

The **General Problem Solver (GPS)** was an early artificial intelligence software program developed in the late 1950s by Allen Newell, Herbert A. Simon, and J. C. Shaw. It was designed to model human problem-solving by treating problems in a general, structured way rather than building specialized solutions for each task. GPS represented a problem as an initial state, a goal state, and a set of possible actions (called operators). It then attempted to find a sequence of steps that would transform the initial state into the goal state, effectively framing intelligence as a process of searching through possible solutions. For instance:

```
Goal: Prove statement X
Difference detected between current state and goal
Select operator: Apply rule A
New subgoal: Establish condition Y
Apply operator: Rule B
Condition Y satisfied
Return to main goal
Apply operator: Rule C
Goal achieved
```

- Lisp: First "AI" Programming Language (1958)

LISP (short for LISt Processing) is one of the earliest programming languages in AI, created in 1958 by John McCarthy. It was designed specifically to handle the kinds of **symbolic manipulation** that early AI researchers believed were at the core of intelligence—things like reasoning, language, and problem solving. Unlike languages focused on numbers and calculations, LISP was built to work with symbols, expressions, and lists, making it ideal for representing knowledge and rules. For example: instead of in Python:

```
In [ ]: def sum_numbers(numbers):
        total = 0
        for number in numbers:
            total += number
```

```
    return total

result = sum_numbers([1, 2, 3, 4, 5])
```

Instead, you have:

```
objects ("dog", "block A")
properties ("red", "on top of")
relationships ("larger than", "is a parent of")
logical statements ("if hungry then eat")
```

- ELIZA (1966)

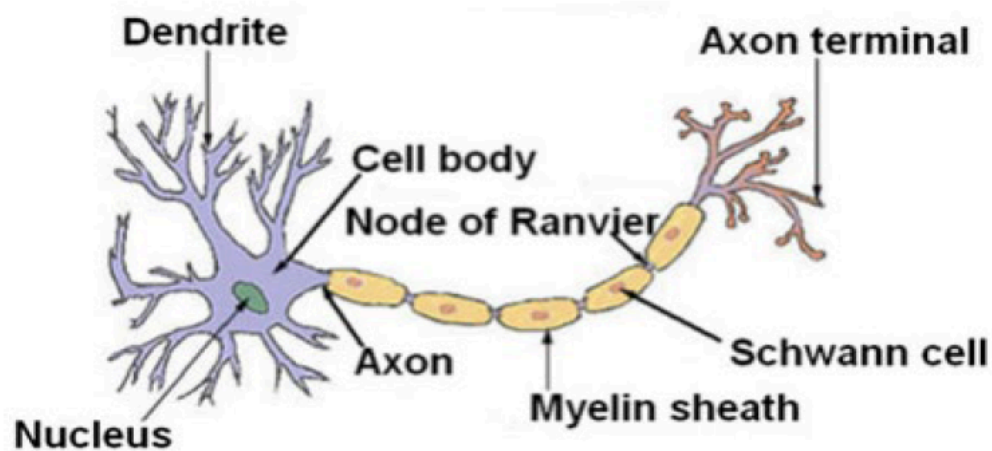
ELIZA was one of the earliest AI programs designed to simulate human conversation. It was created in the mid-1960s by Joseph Weizenbaum at MIT. ELIZA didn't understand language in a human sense; instead, it used pattern matching and simple rules to respond to user input. The most famous version, called DOCTOR, mimicked a Rogerian psychotherapist by turning the user's statements into questions—encouraging them to keep talking.

<https://www.masswerk.at/elizabot/eliza.html>

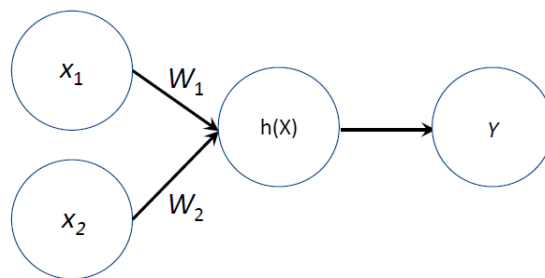
3. Rise of Expert Systems (1970s - 1980s)

Expert systems used rule-based logic to simulate decision-making. They worked best in narrow domains but were not adaptive or scalable. They could not handle uncertainty or learn from new data (unlike modern ML systems).

- MYCIN (Stanford): Diagnosed bacterial infections and recommended treatments.
 - Surprisingly accurate, but never widely used due to legal and trust issues.
- SHRDLU (1972)
 - Created by Terry Winograd.
 - Interacted with users to manipulate virtual blocks (a "block world") using natural language.
 - Demonstrated that AI could understand and follow simple typed commands in a limited environment. <https://patrickvanbergen.com/blocks-world/>
- Neural Networks (1986 - present)
 - Neural networks are software constructs that attempt to mimic neurons in our brains.



Dendrites are the "inputs". The Soma is for "processing". The Axon is the "output signal". This is the inspiration for neural networks—not a literal copy.



$$Y = h(W_1X_1 + W_2X_2)$$

- X are scalar or vector inputs.
- W are scalar or vector weights that represent the strength of a connection.
- $h(x)$ is called an activation function, which decides if neuron sends an output.

Here, the "features". The "weights" specify the importance of features. The summation of features and weights are processed by activation functions and turned in outputs.

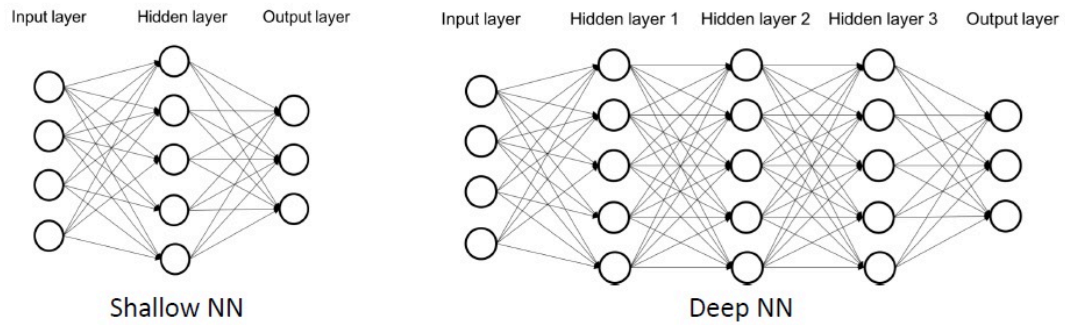
- Machine Learning (1987 - present)

A tool for recognizing patterns and making predictions from data. A neural network is one of many machine learning tools.

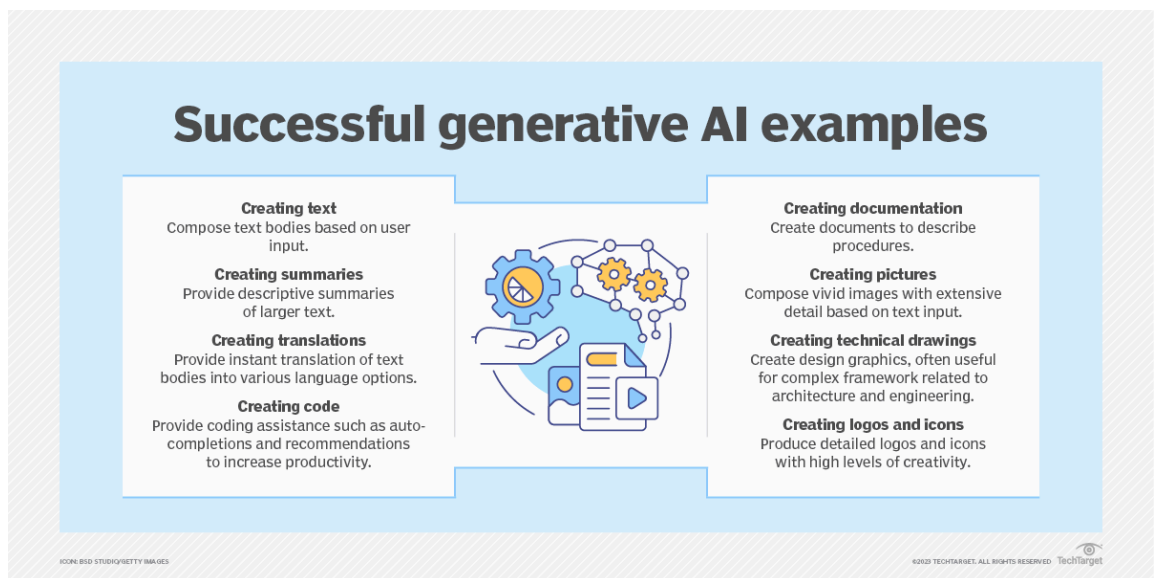
4. Current AI systems

- Deep learning (2011 - present)

Scientists discovered that adding additional layers to neural networks can assist in learning complex patterns.



- Discriminative AI is another machine learning tool that classifies data into **labels**. For instance, an email filter has to decide (i.e. **label**) whether or not an email is spam. It passes the email through a **classifier** and determines whether this email matches a pattern that the classifier understands as either spam or not.
- Generative AI generates **new content**. For example:



5. Cultural Influence of AI in Media

- The 20th century saw an explosion of AI themes in books and films.
- Notable examples:
 - HAL 9000 in 2001: A Space Odyssey (1968): Intelligent but morally ambiguous.
 - R2-D2 & C-3PO in Star Wars (1977): Friendly and functional robots.
 - The Terminator (1984): AI as a destructive force.

These portrayals shaped public hopes and fears, exaggerating AI's capabilities.

<https://www.bbc.com/culture/article/20241017-the-terminator-how-james-camerons-science-fiction-slasher-film-predicted-our-anxieties-about-ai>

6. Challenges & Misconceptions

Early successes and bold claims were made about artificial intelligence in the 1950s and 1960s. Pioneers like Marvin Minsky and Herbert Simon believed that human-level intelligence might be achieved within a **generation**. Early systems could solve algebra problems, prove simple theorems, and play games, which fueled optimism. However, these programs worked well only in constrained settings. When researchers tried to scale them to real-world complexities (vision, language understanding, and common-sense reasoning) they ran into fundamental limitations. As results lagged behind promises, governments and sponsors reduced support, ushering in the first AI **winter** in the 1970s.

The 1980s saw a rise and fall of **expert systems**. These systems encoded human knowledge as rules and were successfully deployed in domains like medical diagnosis and industrial configuration. Companies invested heavily, and AI again seemed poised for widespread impact. But expert systems proved brittle: they were expensive to build, difficult to maintain, and struggled with uncertainty or changing conditions. As costs rose and benefits plateaued, many projects were abandoned. At the same time, specialized AI hardware companies collapsed, and corporate enthusiasm waned. By the early 1990s, funding and hiring had cooled significantly.

Several common factors explain both winters. First, **overpromising** created unrealistic expectations about near-term capabilities. Second, **technical limitations**—insufficient computing power, lack of data, and immature algorithms—made it hard to deliver robust, general solutions. Third, **mismatch between demos and deployment** meant that systems that worked in labs failed in messy real-world environments. When stakeholders realized the gap, investment pulled back.

Importantly, AI winters were not periods of zero progress. Foundational work continued in areas like probability, optimization, and machine learning. These quieter advances eventually enabled later breakthroughs. The resurgence of AI in the 2000s and especially the 2010s—driven by large datasets, improved algorithms, and powerful hardware—shows that winters can be temporary pauses rather than endpoints.

7. Exit Ticket

1. In one or two sentences, what is an AI Winter?
2. What is one reason early AI systems failed to meet expectations?
3. Give one example of discriminative AI.
4. Give one example of generative AI.
5. What is one way AI has influenced culture or media?

8. Lesson Summary: AI in the 20th Century

Artificial intelligence began with great optimism in the 1950s and 1960s, as early researchers believed machines would soon be able to match human reasoning. Important figures like Alan Turing and John McCarthy helped lay the foundation, and early programs such as ELIZA and the General Problem Solver demonstrated that computers could mimic certain aspects of thinking. However, these systems worked only in limited situations and could not handle the complexity of the real world.

This gap between expectations and reality led to periods known as **AI Winters**, when funding and interest declined because AI systems failed to deliver on bold promises. As described in your lecture, these cycles were driven by **overpromising, technical limitations, and systems that worked in controlled demos but failed in practical use**. A similar pattern occurred again in the 1980s with expert systems, which were useful but ultimately too rigid, expensive, and difficult to maintain.

Despite these setbacks, AI continued to evolve. Neural networks and machine learning introduced new ways for computers to learn from data rather than relying solely on fixed rules. This eventually led to modern advances like deep learning, which uses multiple layers to recognize complex patterns.

The lesson also highlights two important types of AI used today. **Discriminative AI** classifies data into categories, such as identifying whether an email is spam. **Generative AI**, on the other hand, creates new content, such as generating text or images. Understanding this distinction helps explain many modern AI tools.

Finally, AI has had a strong influence on culture and media. Films and stories have shaped how people think about AI—sometimes creating excitement, and other times fear—often exaggerating what AI can actually do.

Overall, the history of AI shows a repeating pattern: excitement, disappointment, and progress. Today's AI is more successful because of better data, stronger computing power, and improved techniques, but the lessons from the past remind us to stay realistic about what AI can achieve.

8. Vocabulary

Vocabulary (10 Terms with Definitions)

1. **AI Winter** – A period when excitement and funding for AI collapse because systems fail to deliver on the promises made by researchers or the media.

2. **SNARC** – An early (1950) reinforcement-learning machine built by Marvin Minsky and Dean Edmonds from vacuum tubes and analog components to simulate a rat learning to navigate a maze.
3. **Reinforcement Learning** – A way of learning through trial and error, where successful actions are reinforced over time — the core idea introduced by SNARC decades before it became a major field in AI.
4. **General Problem Solver (GPS)** – An early AI program by Newell, Simon, and Shaw that modeled problem-solving as searching from an initial state to a goal state using a set of operators.
5. **LISP** – A programming language created by John McCarthy in 1958, designed to represent symbols, objects, properties, and rules rather than just numbers.
6. **ELIZA** – An early natural-language program created by Joseph Weizenbaum that used pattern matching, not true understanding, to simulate conversation — most famously as the “DOCTOR” script.
7. **Expert System** – A rule-based AI program that simulates expert decision-making in a narrow domain, such as MYCIN’s bacterial-infection diagnoses, but cannot adapt, scale, or handle uncertainty.
8. **Neural Network** – A software system loosely inspired by neurons in the brain, where inputs, weights, and an activation function combine to produce an output.
9. **Discriminative AI** – AI that classifies or labels data, such as deciding whether an email is spam or not.
10. **Generative AI** – AI that creates new content, such as text, images, or music.